Simulating the Classical Distributions

A working tour from uniform bits to the t-test,
with the `probsim` C99 library

The `probsim` project

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Abstract

This paper accompanies `probsim`, a small C99 library that simulates the thirteen classical probability distributions — plus a three-member annex for modeling cryptographic hashes — and carries the results all the way to working t-tests. For each distribution we explain what it models, where it earns its keep in statistical inference and applied modeling, and how the library simulates it — always with the simplest algorithm that is still practical. The distributions are presented as one connected family (Figure 1): every sampler is built from uniform bits, and the family funnels naturally into Student's t, whose tail areas turn simulated or observed data into p-values. Every section names the source file and function that implements it, and the sources cite these sections back, so paper and code can be read side by side. This paper also has an interactive web edition (`docs/index.html`, served via GitHub Pages) with every figure animated — each figure caption below links to its interactive counterpart. A sibling project, randtests, surveys the statistical test suites (Diehard, NIST SP 800-22, TestU01, PractRand) that put several of these distributions to work against real generators and hash functions. A second sibling, `shastats` (The `shastats` project, 2026), hashed eight and a half billion messages to test whether SHA-256's leading-zero counts obey the laws developed here; its measurements recur through the paper as worked field applications of the binomial (§3.2), geometric (§3.3), uniform (§4.1), chi-squared (§4.6), Gumbel (§4.10) and dispersion (§5.4) machinery.

The distributions at a glance

1 Introduction

A probability distribution is a reusable answer to the question "what does this kind of randomness look like?" A handful of them — the *classical* distributions — cover a remarkable share of practical work: coin-like events, counts, waiting times, measurement noise, and the sampling behaviour of statistics computed from data. Learning them as isolated formulas is hard; learning them as a *family*, where each new member is a short construction from ones you already know, is much easier. That family view is exactly how the `probsim` library is written, and Figure 1 is its map.

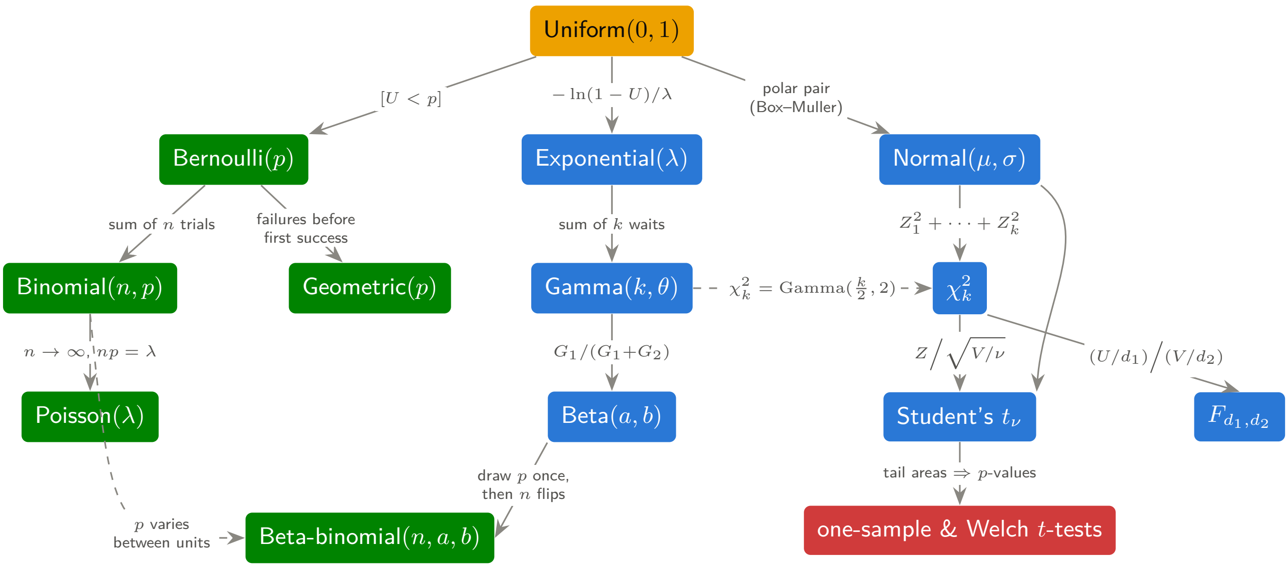

Figure 1: The classical family as `probsim` builds it. Everything starts from uniform bits (§2); discrete distributions grow along the left (§3), continuous ones along the right (§4), and the family funnels into the t-tests of §5. Each arrow is literally a line or two of C in `src/dist.c`. The beta-binomial (§3.6) is the one arrow that runs *back* from the continuous side to the discrete: it is the binomial with its p drawn afresh from a beta for each unit rather than fixed once for all. Three further relatives for hash modeling hang off the tree: the negative binomial (§3.5) is a sum of geometric waits, and the Rayleigh and Gumbel (§4.9–4.10) are one-line transforms of the exponential. [interactive version on the web]

How to read this with the code. The library is four small C files behind one header, `include/probsim.h`. Section §2 covers `src/rng.c`; §3 and §4 cover `src/dist.c`; §5 covers `src/special.c` and `src/stats.c`; §6 walks through the driver `app/simulate.c` and the test suite `tests/test_probsim.c`. Each subsection below ends with a pointer to its implementation, and the source comments point back here. Throughout, the guiding rule is the one this library was written under: *prefer the simplest algorithm that is easy to understand and still practical to use.* Where a cleverer method is the industry standard — the ziggurat for normals, say (Marsaglia and Tsang, 2000b) — we cite it, explain the trade-off, and keep the simple one.

2 Where the randomness comes from

Computers are deterministic, so "random" numbers are produced by a *pseudo-random number generator* (PRNG): a small state machine whose output sequence is indistinguishable, for statistical purposes, from true randomness. `probsim` uses **xoshiro256++** (Blackman and Vigna, 2021): 256 bits of state, a period of $2^{256} - 1$, excellent results on the standard test batteries, and a step function of about ten lines — comfortably inside our simplicity rule. Reasonable alternatives with the same spirit include PCG (O'Neill, 2014); the deeper theory of testing generators is in Knuth (1997, ch. 3). None of these are cryptographic generators, and `probsim` should never be used where an adversary matters.

Two practical details deserve attention because they bite working statisticians. First, *seeding*: naive seeds like $1, 2, 3$ are nearly identical bit patterns, and a weak seeding routine would start the streams in correlated states. The library therefore runs every seed through the splitmix64 mixer, as the xoshiro authors recommend. Second, *explicit state*: every sampler takes a `ps_rng *` argument and there is no hidden global, so a simulation is reproducible from a single recorded seed and independent streams are just independent structs — which is what makes the test suite of §6 deterministic.

2.1 Uniform is enough

The raw generator emits 64-bit integers; dividing the top 53 bits by 2^{53} gives a double U uniform on $[0, 1)$ at full mantissa precision. Everything else in this paper is manufactured from such U's. The master tool is *inverse-transform sampling*: if F is a CDF and $U \sim \text{Uniform}(0, 1)$, then $X = F^{-1}(U)$ has CDF exactly F, because $\Pr[X \le x] = \Pr[U \le F(x)] = F(x)$. Figure 2 shows the idea; the exponential sampler of §4.2 is its purest one-line application. When F^{-1} is awkward, two other levers — *transformation* (build the variable from simpler ones) and *rejection* (propose, then accept with the right probability) — take over; both appear in §4. The encyclopedia of all three techniques is Devroye (1986).

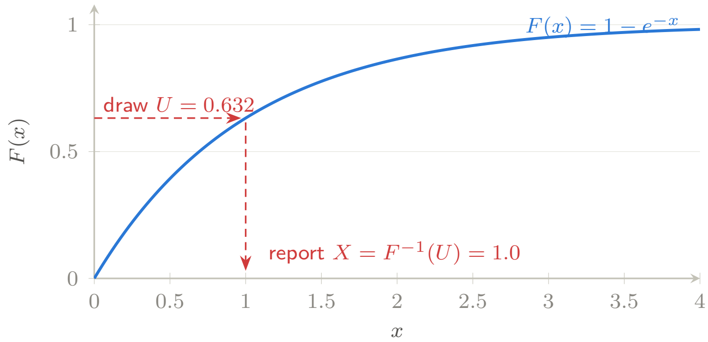

Figure 2: Inverse-transform sampling: a uniform draw on the vertical axis is pushed through F^{-1} to land on the horizontal axis with density exactly F'. Regions where F rises steeply receive more of the uniform mass — that is the whole trick. [interactive version on the web]

↪ implemented in `src/rng.c`: `ps_rng_seed()`, `ps_rng_next()`, `ps_runif()`

2.2 The generator under the microscope

Since every result in this library is only as trustworthy as its stream of uniforms, the generator deserves a closer look: what kind of object xoshiro256++ is, which of its properties are *theorems*, which rest on *measurement*, and what remains open.

What it is. xoshiro256++ belongs to the family of *scrambled linear* generators (Blackman and Vigna, 2021): a linear *engine* advances the 256-bit state $s = (s_0, s_1, s_2, s_3)$ using only XORs, shifts and rotations — a linear map over the two-element field $\mathbb{F}_2$ — and a small nonlinear *scrambler*, here $\text{rotl}(s_0 + s_3, 23) + s_0$ with 64-bit addition, hashes the state into each output word (Figure 3). The division of labour is deliberate. Linear maps are the one class of generators whose long-run structure mathematics can fully analyse, but their raw outputs carry visible linear fingerprints; the scrambler destroys those artifacts while provably not shortening the cycle, since it never feeds back into the state. The design refines a lineage that begins with Marsaglia's xorshift generators (Marsaglia, 2003) and their systematic re-examination in Vigna (2016).

What is proved. The engine's transition matrix M over $\mathbb{F}_2$ has a *primitive* characteristic polynomial $P(x) = \det(xI - M)$ — degree 256, weight 115 (an odd weight, as any irreducible binary polynomial must have). Packing bit i with the coefficient of x^i, it reads, in hexadecimal,

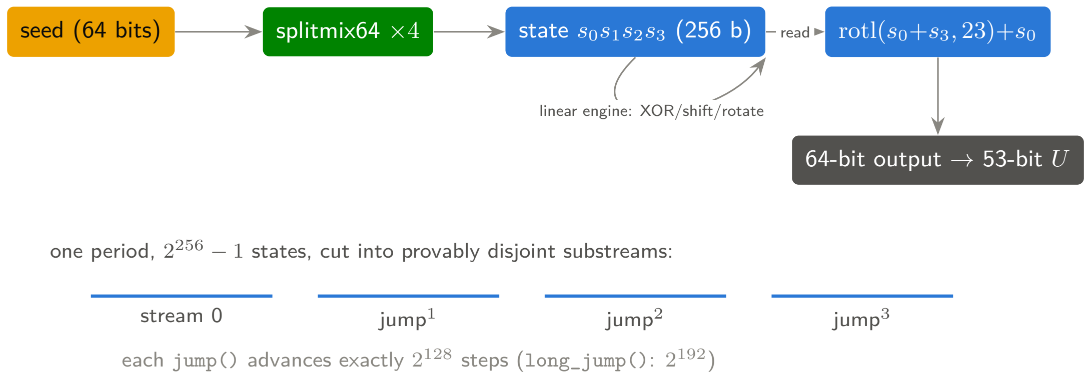

Figure 3: Anatomy of xoshiro256++: a seed is spread over 256 bits of state by splitmix64; an $\mathbb{F}_2$-linear engine walks a single cycle of length $2^{256}-1$; a nonlinear scrambler hashes each state into the output word whose top 53 bits become the uniform draw of §2.1. Jump functions cut the cycle into provably non-overlapping substreams for parallel simulation. [interactive version on the web]

`0x1 0003c03c 3f3ecb19 04b4edcf 26259f85 0280002b cefd1a5e 9d116f2b b0f0f001.`

The repository does not ask the reader to take this on faith: `scripts/charpoly.py` rederives P from the engine code itself (Berlekamp–Massey on an output-bit sequence, certified against a second independent state bit) and then proves primitivity by verifying $x^{2^{256}-1} \equiv 1$ and $x^{(2^{256}-1)/q} \not\equiv 1 \pmod{P}$ for every prime q dividing $2^{256}-1$, whose complete factorization is classical (it descends from the Fermat numbers F_0–F_7). On the algebra of primitive polynomials and linear recurrences see Lidl and Niederreiter (1997, ch. 3, ch. 8); the design context is Blackman and Vigna (2021). Primitivity is exactly the central theorem: the state sequence is one single cycle through *every* nonzero 256-bit state, so the period is exactly $2^{256}-1$. Two corollaries matter in practice. First, the all-zero state is the unique fixed point and must never be used — one reason `ps_rng_seed()` runs every seed through splitmix64 rather than loading it directly. Second, because the engine is linear, jumping ahead by any fixed number of steps is a matrix power, computable exactly: the published `jump` and `long-jump` polynomials advance the state by precisely 2^{128} and 2^{192} steps, giving parallel simulations substreams that are *guaranteed* never to overlap. On the output side, the authors use the theory of filtered linear-feedback shift registers to prove properties of the scramblers themselves, and for the sibling multiplicative scramblers ($*$ and $**$) they prove $n/64$-dimensional equidistribution — four dimensions at this state size: every quadruple of consecutive outputs appears equally often over the full period.

What is measured. No practical non-cryptographic generator comes with a proof that it passes statistical tests, so the case for quality is empirical — and should be read the same way as any experiment. xoshiro256++ shows *no systematic failures* on BigCrush, the most demanding battery in the TestU01 suite (L'Ecuyer and Simard, 2007), and no anomalies in long PractRand runs (Doty-Humphrey, 2014). (What these batteries actually contain, and how they compare with the rest of the testing landscape, is the subject of the sibling survey randtests.) More telling is a test the authors built precisely because the standard suites missed something: raw $\mathbb{F}_2$-linear engines exhibit long-range dependencies in the *Hamming weight* (popcount) of their outputs, invisible to BigCrush. Their Hamming-weight dependency test detects such bias in unscrambled and weakly scrambled generators, while xoshiro256++ passes it in the tested ranges (Blackman and Vigna, 2021) — the empirical confirmation that the scrambler is doing its job. The contrast with the family's own history is instructive: the low-order bits of the simpler xoshiro256+ inherit low linear complexity from the engine and fail linearity-sensitive tests, which is exactly why this library uses the ++ scrambler.

What is open, and what is simply false. Openness, first: there is no theorem that xoshiro256++ passes any statistical test — for these generators such guarantees are structural (period, equidistribution, jumps) while test-passing is evidence, bounded by the compute anyone has spent; the $n/64$ equidistribution theorem is stated for the multiplicative scramblers, not for ++, whose output equidistribution is not fully characterised; and a battery that fails it tomorrow is always possible in principle — that is the nature of empirical quality. Falsity, second: xoshiro256++ is *not* a cryptographic generator. Its state-recovery resistance is nil by design — observing a modest number of outputs suffices to reconstruct the state and predict the stream — so it must never guard secrets, tokens or shuffles an adversary can bet against. For simulation, the working summary is comfortable: everything this paper's simulations need — reproducibility, stream independence, full-period coverage, absence of detectable bias at our sample sizes — is either proved or measured with margin to spare.

↪ implemented in `src/rng.c` `(engine, scrambler and seeding, with these properties cited)`

3 The discrete family

Discrete distributions describe counts. The four classical ones form a single story: a yes/no event (Bernoulli), how many yeses in n tries (binomial), how long until the first yes (geometric), and how many events occur when opportunities are numerous but each is rare (Poisson). The driver `app/simulate.c` exercises each with the worked parameters used below.

3.1 Bernoulli

The Bernoulli(p) variable is a single yes/no experiment: $\Pr[X{=}1] = p$, $\Pr[X{=}0] = 1 - p$, with $\mathbb{E}X = p$ and $\operatorname{Var} X = p(1-p)$. It is the atom of applied statistics: a conversion in an A/B test, a defective part, a patient responding to treatment, a loan defaulting. Almost all of *binary-outcome* inference — estimating a proportion, comparing two proportions, logistic regression — is Bernoulli modeling. Simulation is the definition itself: draw U and report whether $U < p$.

Origins and habitat. The distribution carries the name of Jacob Bernoulli, whose posthumous *Ars Conjectandi* (Bernoulli, 1713) proved the first law of large numbers — his "golden theorem" that the observed frequency of successes converges to p — and thereby turned the coin toss from a gambling prop into the founding object of mathematical statistics. The model arises naturally wherever an outcome is *recorded* as a dichotomy, which is a choice as much as a fact of nature: cure/no cure, click/no click, default/no default. Its independence-plus-constant-p idealisation is what everything downstream inherits, so the practical craft lies in checking it — runs of correlated outcomes or drifting p break the binomial counts and waiting times built on top. In modern practice the Bernoulli likelihood is the atom of logistic regression, of web-scale A/B platforms running thousands of simultaneous coin-flip experiments, and of the Beta–Bernoulli machinery of §4.5.

↪ implemented in `src/dist.c`: `ps_bernoulli()`

3.2 Binomial

The Binomial(n, p) variable counts successes in n independent Bernoulli(p) trials:

$$\Pr[X = k] = \binom{n}{k} p^k (1-p)^{n-k}, \qquad \mathbb{E}X = np, \quad \operatorname{Var} X = np(1-p).$$

It models any fixed-quota count: heads in n tosses, positive responses among n patients, defects in a batch of n (the basis of industrial *acceptance sampling*), conversions among n visitors. In inference it is the exact sampling distribution behind every test of a proportion — the sign test, the exact binomial test, and, through the normal approximation, the familiar z-test for proportions. `probsim` simulates it as what it is — n Bernoulli draws summed — an $O(n)$ loop that cannot be wrong by inspection; sub-linear samplers exist (Devroye, 1986, ch. X.4) but buy speed with complexity we do not need. Figure 4 (left) shows the worked example $\text{Binomial}(20, 0.25)$.

Origins and habitat. The binomial is probability theory's birth certificate. The 1654 Pascal–Fermat correspondence on the *problem of points* — how to split the stakes of an interrupted game — amounted to computing binomial probabilities from Pascal's triangle, and Bernoulli (1713) supplied the general trials framework. Most consequentially, de Moivre (1738) showed that for large n the binomial is approximated by the curve e^{-x^2}: the first central limit theorem, and the historical doorway through which the normal distribution entered statistics (§4.3). The model arises wherever a fixed number of like units each independently succeed or fail: Mendel's famous 3:1 pea ratios are binomial counts; industrial *acceptance sampling* (developed by Dodge and Romig at Bell Laboratories in the 1920s–40s) decides a lot's fate from the binomial behaviour of a small inspected sample; election audits and A/B tests are the same calculation with different stakes. The standard failure mode is *overdispersion* — p varying between units inflates the variance beyond $np(1-p)$ — which is the empirical signature that the Beta–binomial mixture of §4.5 fits better.

A field measurement: the avalanche of SHA-256. The sibling experiment `shastats` (The `shastats` project, 2026) — a local repository whose measurements recur through this paper — uses the binomial as the yardstick for a cryptographic design goal. SHA-256 is the standard hash function producing a 256-bit *digest*, and the *strict avalanche criterion* demands that flipping one input bit flip each output bit independently with probability $\frac{1}{2}$; the *number* of output bits that change should therefore be $\text{Binomial}(256, \frac{1}{2})$: mean $np = 128$, standard deviation $\sqrt{np(1-p)} = \sqrt{256/4} = 8$. Measured exhaustively over 419,430,400 single-bit-flip pairs, the mean came out 128.0000 and the spread 8.0003. At a finer grain, each of the $256 \times 25 = 6{,}400$ (input bit, output bit) cells is one coin whose null is an *exact* binomial — one of the two places that project trusts an analytic p-value, precisely because the null needs no approximation. The most lopsided cell ran at 0.500475 heads, +3.9 standard errors, which sounds alarming until the maximum of 6,400 readings is priced at its own distribution (§4.10: about 4.35σ expected). The binomial supplied both the prediction and the test that it holds.
↪ implemented in `src/dist.c`: `ps_binomial()`, `ps_binomial_pmf()`

3.3 Geometric

The $\text{Geometric}(p)$ variable counts *failures before the first success* (support $0, 1, 2, \ldots$, the convention of R's `rgeom`): $\Pr[X = k] = p(1-p)^k$, $\mathbb{E}X = (1-p)/p$, $\operatorname{Var} X = (1-p)/p^2$. It is the discrete waiting time: retries before a packet gets through, cold calls before a sale, coin flips before the first head. Its *memoryless* property (past failures tell you nothing about the future) makes it the discrete twin of the exponential — and indeed the sampler exploits exactly that: invert the geometric CDF with one uniform, $X = \lfloor \ln(1-U)/\ln(1-p) \rfloor$, which is the exponential inversion of §4.2 rounded down to a count.

Origins and habitat. "How many throws until the first six?" is as old as dice questions get — de Méré's puzzles that provoked the Pascal–Fermat correspondence were of exactly this waiting-time kind. Two properties give the geometric its character. It is provably the *only* discrete distribution with the memoryless property, so it arises, exactly, whenever a repeated trial has a constant per-attempt hazard; and the everyday refusal to believe this — "red is due" after a run of black — is the gambler's fallacy, which makes the geometric the distribution most humans mis-intuit. It appears naturally as retransmission counts in randomized network protocols (the backoff analyses behind ALOHA and Ethernet), as cycles-to-conception in demography's fecundability models, as the first defect's position in sequential inspection, and as discrete-time survival with constant hazard — the null model against which real ageing or learning effects are measured.

A field measurement: leading zeros of SHA-256. If a hash digest behaves like 256 fair coin flips (§3.2 introduced the sibling experiment `shastats`; §4.9 develops the ideal-hash model), its count of *leading zero bits* is the count of tails before the first head: Geometric($\frac{1}{2}$) in exactly this section's convention, $\Pr[N = k] = 2^{-(k+1)}$, mean $(1 - p)/p = 1$. That count is the quantity Bitcoin's proof of work thresholds — mining (§3.5) is waiting for a digest with enough leading zeros, a geometric race run billions of times per second. `shastats` (The `shastats` project, 2026) put the law itself on trial: 2^{33} messages hashed and histogrammed, then compared against $2^{-(k+1)}$ with *no parameter fitted.* The law holds — goodness of fit $p = 0.30$ (§4.6 shows the χ^2 machinery), sample mean 1.000011 against the theoretical 1 — eight and a half billion digests behaving like the memoryless coin the model promises. The same project's per-population *maxima*, 22 to 33 leading zeros, are priced in §4.10.

↪ implemented in `src/dist.c`: `ps_geometric()`, `ps_geometric_pmf()`

3.4 Poisson

The Poisson(λ) variable is the law of rare events: the limit of Binomial$(n, \lambda/n)$ as $n \to \infty$,

$$\Pr[X = k] = e^{-\lambda}\frac{\lambda^k}{k!}, \qquad \mathbb{E}X = \operatorname{Var} X = \lambda.$$

It models counts per unit of exposure when events are individually unlikely but opportunities are many: arrivals at a queue per minute, radioactive decays per second, typos per page, insurance claims per policy-year, sequencing reads per gene. In inference it anchors count regression (Poisson GLMs, with the $\mathbb{E}X = \operatorname{Var} X$ property serving as the canonical check for *overdispersion*) and classical goodness-of-fit tests. The library simulates it with Knuth's product-of-uniforms method (Knuth, 1997, §3.4.1) — count uniforms until their running product drops below $e^{-\lambda}$ — applied in chunks of rate at most 30 so the product never underflows; the chunking is valid because Poisson rates add. Its CDF ships through the *continuous* family: $\Pr[N \le k] = 1 - P(k+1, \lambda)$ with P the regularized incomplete gamma of §5.1, because the count of events by time λ and the wait for the $(k+1)$-th event are two descriptions of one process — a discrete/continuous duality developed fully at the chi-squared (§4.6). Figure 4 (right) shows Poisson(4).

Origins and habitat. Poisson derived his limit in an 1837 treatise on the probability of criminal-jury verdicts, where it went largely unnoticed; it was von Bortkiewicz (1898), with his "law of small numbers", who made it famous — his tabulation of deaths by horse kick across Prussian cavalry corps remains the canonical demonstration that rare-event counts follow the Poisson law with uncanny fidelity. Two independent rediscoveries cemented its physical and industrial standing: Rutherford and Geiger's 1910 alpha-particle counts matched Poisson predictions and helped establish the randomness of radioactive decay, while Erlang's 1909 analysis of telephone-call arrivals at the Copenhagen exchange founded teletraffic theory and, with it, queueing. The genesis condition is superposition: many independent sources, each unlikely to fire in any short interval, add up to Poisson counts — which is why the model recurs for insurance claim frequencies, mutations along a genome, web-server request arrivals, and photons at a detector. As with the binomial, real data often show variance exceeding the mean; that overdispersion, and the gamma-mixed Poisson (negative binomial) that absorbs it, is the standard second chapter of count modeling.

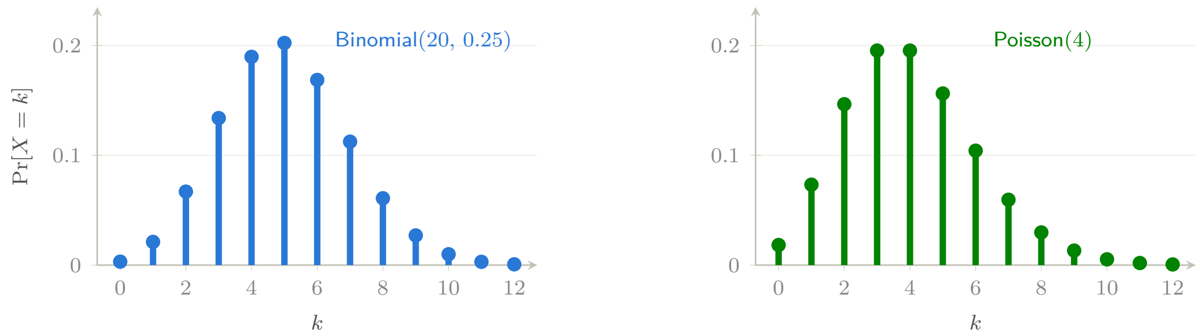

Figure 4: The two classical count laws, at the driver's worked parameters. Both have mean $\approx$ 4–5, but the binomial is confined to $\{0, \ldots, 20\}$ while the Poisson has an unbounded right tail — the visible signature of "many opportunities, each rare". [interactive version on the web]

↪ implemented in `src/dist.c`: `ps_poisson()`, `ps_poisson_pmf()`, `ps_poisson_cdf()`

3.5 Negative binomial

The negative binomial $\mathrm{NB}(r, p)$ counts *failures before the r-th success*:

$$\Pr[X = k] = \binom{k+r-1}{k} p^r (1-p)^k, \qquad \mathbb{E}X = \frac{r(1-p)}{p}, \quad \mathrm{Var}\, X = \frac{r(1-p)}{p^2},$$

the geometric of §3.3 being the case $r = 1$ — and the sampler is that identity, a sum of r geometric waits. Its CDF is one more incomplete-beta reduction, $\Pr[X \le k] = I_p(r,\, k+1)$ (§5.1). It wears two very different hats. As a *waiting time* it prices repeated search: if each independent attempt succeeds with probability p, the total work to collect r successes is NB-distributed — which, with p of order 2^{-n}, is exactly the cost model for finding r preimages or r collisions against an ideal n-bit hash, the multi-target attacks of §4.9's annex (the empirical side of that story — collision statistics as a working test of real hash functions — is covered by the sibling survey). As a *count model* it is the gamma-mixed Poisson: let the Poisson rate of §3.4 itself vary as a gamma (§4.4) across units and the marginal count is negative binomial, with the variance exceeding the mean — the promised cure for overdispersion.

Origins and habitat. The pmf appears (as a gambling wait) in Montmort's 1714 analysis of the problem of points, and the name records a formal joke: the coefficients come from the binomial series with a *negative* exponent. Its modern statistical career began with Greenwood and Yule (1920), who derived it as the law of accident counts among London munitions workers when individual proneness varies — the original overdispersion argument, gamma-mixing included — and it has been the default model for clumped counts ever since: insurance claims, species abundance in quadrats, crime counts, and today the per-gene read counts of RNA-seq, whose standard analysis tools are negative-binomial regressions at scale.

Worked example: the Bitcoin double-spend race. The two hats meet on one head in the security argument of the Bitcoin white paper (Nakamoto, 2008). A merchant waits for z confirmations — z further blocks on the honest chain — before releasing goods; an attacker holding a fraction q of the network's hash power mines a competing chain in secret, hoping to overtake it and undo the payment. Blocks are found by a Poisson process (§3.4), so by thinning each block belongs to the attacker with probability q and to the honest network with probability $p = 1 - q$, independently. Then the number K of blocks the attacker finds while the honest chain advances by z is *exactly* $\mathrm{NB}(z, p)$: attacker blocks are the failures before the z-th honest success. Take the other route and the other hat appears: the honest network needs an Erlang — $\mathrm{Gamma}(z,\ 1/p\lambda)$, §4.4 — amount of time to find z blocks, and given that time the attacker's count is Poisson, so K is a gamma-mixed Poisson. Same law, both derivations.

The white paper takes a third route: it replaces that random elapsed time by its mean, and so arrives at a Poisson count with $\lambda = zq/p$. The mean is right — $\mathbb{E}K = zq/p$ — but the dispersion is not. $\mathrm{NB}(z, p)$ has variance zq/p^2, which is $1/p$ times its mean, so the Poisson understates the variance by that factor: 11% at $q = 0.1$, 67% at $q = 0.4$. The slip is not in any cryptographic assumption; it is the familiar one of substituting a mean into a non-linear function, and it happens to understate exactly the tail the merchant is buying protection against. Rosenfeld (2014) corrected it — his paper declines the assumption and models the count "more accurately as a negative binomial variable" — and Grunspan and Pérez-Marco (2018) give the exact success probability in closed form — one more incomplete-beta reduction (§5.1):

$$P(z) = I_{4pq}\left(z, \tfrac{1}{2}\right).$$

So the true risk shrinks by a factor $4pq$ for each further confirmation, while the Poisson version shrinks by e^{-c} with $c = q/p - \ln(q/p) - 1$, the large-deviation rate of a Poisson count — strictly faster: 0.27 per confirmation against 0.36 at $q = 0.1$. Two different decay constants is the signature of two different models, and the gap compounds with every confirmation the merchant waits: at $q = 0.1$ and $z = 10$ the real risk is 7.9×10^{-6} against the 1.2×10^{-6} assumed, a factor of 6, rising to a factor of 84 by $z = 20$ (Figure 5).

This example is also a small lesson in why the library keeps the incomplete beta of §5.1 rather than summing series. The textbook route to $P(z)$ — sum $\Pr[K = k]$ times the gambler's-ruin catch-up probability over $k < z$, then subtract from 1 — is a subtraction of two quantities that agree to more and more digits as z grows. By $z = 40$ at $q = 0.1$ the true answer, 2.0×10^{-19}, is far below the rounding error of that final subtraction, and in double precision the sum returns noise: this build reports 5.9×10^{-15} there, and a negative probability at $z = 35$. The closed form `ps_incbeta(4pq, z, 0.5)` returns the same $z = 40$ figure to fifteen digits. `tests/test_probsim.c` checks the two forms against each other over the range where both are trustworthy, pins the breakdown, and checks that the Poisson version understates the risk.

↪ implemented in `src/dist.c`: `ps_negative_binomial()`, `ps_negative_binomial_pmf()`, `ps_negative_binomial_cdf()`; `src/special.c`: `ps_incbeta()`

3.6 Beta-binomial

Every distribution so far flips one coin. The beta-binomial draws the coin itself: a proportion $p \sim \mathrm{Beta}(a, b)$ (§4.5), once per unit, and only then n flips of *that* coin. Integrating p out leaves

$$\Pr[X = k] = \binom{n}{k} \frac{B(k + a,\ n - k + b)}{B(a, b)}, \qquad \mathbb{E}X = n\bar{p}, \quad \mathrm{Var}\,X = n\bar{p}(1 - \bar{p})\big[1 + (n - 1)\rho\big],$$

with $\bar{p} = a/(a{+}b)$ and $\rho = 1/(a{+}b{+}1)$. The mean is the binomial's; the variance is the binomial's inflated by $1 + (n - 1)\rho$. That factor is the whole distribution. Its ρ is the *intraclass correlation* — the chance that two draws from the same unit agree beyond what the common rate explains — and it interpolates between the two experiments people habitually confuse: $\rho \to 0$ sends $a{+}b \to \infty$, the beta collapses onto $\bar{p}$, and the beta-binomial *is* $\mathrm{Binomial}(n, \bar{p})$; $\rho \to 1$ gives a coin that is all heads or all tails per unit. Note where the mixing happens. Drawing a fresh

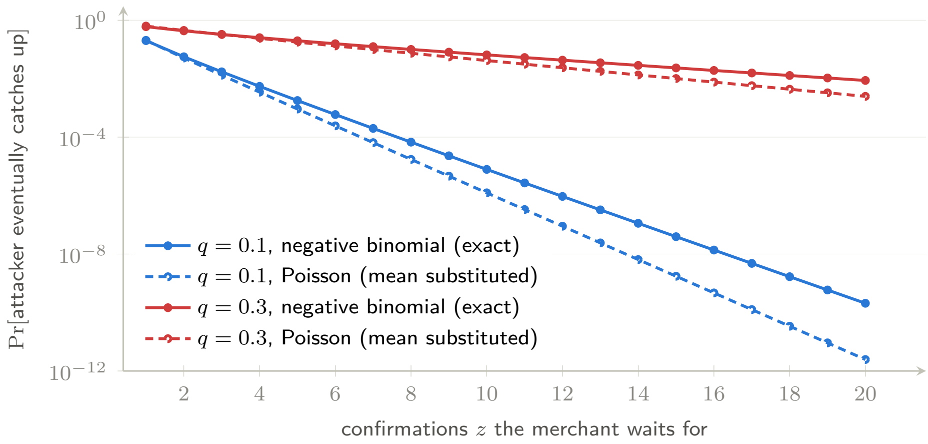

Figure 5: The cost of the wrong count law, on a log scale. Solid curves are the exact negative-binomial risk $I_{4pq}(z, \frac{1}{2})$; dashed curves of the same colour are the Poisson approximation obtained by substituting the mean elapsed time, at the same attacker hash-power share q. The two agree for the first confirmation or two — which is why the substitution survived scrutiny — and then separate geometrically, because they decay at different rates. Waiting longer buys less safety than the Poisson calculation promises. [interactive version on the web]

p for every *flip* would give the plain binomial straight back; drawing it once per *unit* is what correlates the flips within a unit, and the library's sampler is that sentence in two lines — one `ps_beta`, then one `ps_binomial`.

This is the binomial's overdispersion cure exactly as the negative binomial (§3.5) is the Poisson's, and the parameterisation worth using in practice is $(\bar{p}, \rho)$ rather than (a, b): `ps_beta_binomial_ab()` inverts $s = 1/\rho - 1$, $(a, b) = (\bar{p}s, (1-\bar{p})s)$. It earns its keep wherever trials cluster and the cluster has its own propensity: litters, couples and plots in biology, production lots in acceptance sampling, classrooms and clinics in hierarchical models, and — via the empirical-Bayes reading of the same two-stage story — the shrinkage estimators that pull small lots' noisy rates toward the population mean. The CDF has no closed form (its tail is a ${}_3F_2$), so the library sums the pmf from whichever end is shorter.

Origins and habitat. The two-stage picture is as old as Bayes (1763): his billiard ball fixes an unknown p before any subsequent ball is thrown, which makes the resulting predictive law for k successes in n the first beta-binomial in print. It became a working tool with Skellam (1948), whose title states the construction outright — the binomial, with the probability of success regarded as variable between the sets of trials. Its most persistent habitat is the question of whether boys or girls run in families, and that literature is a lesson in what ρ costs to see. Geissler's 1889 registry of Saxon births, about a million families, is the canonical test bed: Edwards (1958) found the binomial fit genuinely wanting, and Lindsey and Altham (1998) later fitted Skellam's mixture and its rivals to the same table, finding the dispersion real but small and entangled with family size — larger families show both a higher Pr[boy] and more dispersion, which is what a stopping rule (keep trying for a son) would also produce. Against that, Jacobsen et al. (1999) took 815,891 Danish births and found *no* independent effect of the sexes of the preceding children. The disagreement is not a contradiction so much as an illustration: the heterogeneity at issue is of order $\rho \sim 10^{-3}$, which is invisible in any ordinary study, detectable only at registry scale, and confounded there by exactly the family-level behaviour it resembles. §5.4 makes that arithmetic explicit, and §6.2 runs it.

↪ implemented in `src/dist.c`: `ps_beta_binomial()`, `ps_beta_binomial_pmf()`, `ps_beta_binomial_cdf()`, `ps_beta_binomial_ab()`

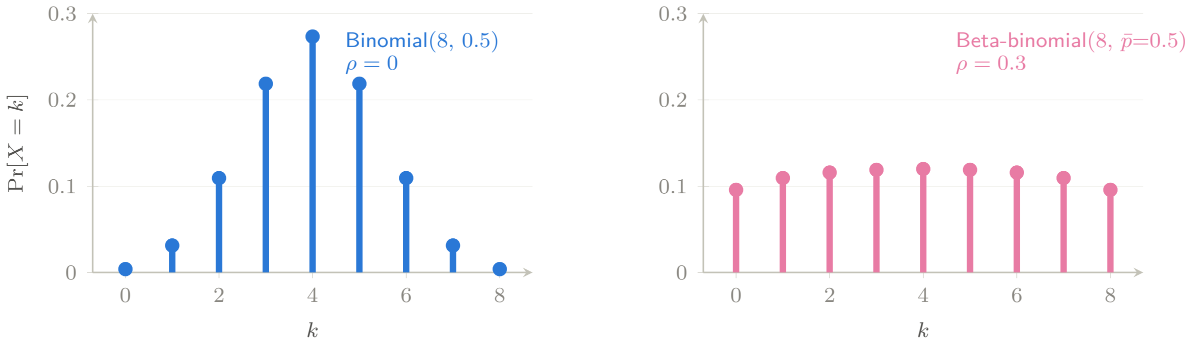

Figure 6: The same eight flips at the same mean rate, differing only in where the coin comes from. Left: one coin for everyone — the binomial concentrates on four heads. Right: a coin drawn per unit with intraclass correlation $\rho = 0.3$ — the mean is unmoved at 4, but the mass has fled to the extremes, because a unit that drew a high p tends to produce a run. The variance ratio is $1 + (n-1)\rho = 3.1$. [interactive version on the web]

4 The continuous family

Continuous distributions describe measurements. We meet them in the order the library constructs them (the right-hand side of Figure 1), because each is a short recipe from the previous ones.

4.1 Uniform

Uniform(a, b) spreads its mass evenly: density $1/(b-a)$ on $[a, b)$, mean $(a+b)/2$, variance $(b-a)^2/12$. Beyond being the raw material of §2.1, it is a model in its own right (rounding error, phases, arrival offsets) and a cornerstone of inference through one beautiful fact: *under a true null hypothesis, a p-value is Uniform*$(0, 1)$. That fact is testable — the test suite verifies it empirically in §6.3 — and it underlies p-value histograms, false-discovery-rate methods, and simulation-based calibration. Simulation is one affine map of U.

A field measurement: 256 p-values on trial. The sibling experiment `shastats` (§3.2; The `shastats` project, 2026) leans on p-value uniformity as a working instrument. Its sweep asks whether prefixing a byte to a message moves SHA-256's leading-zero distribution (§3.3); each of the 256 possible prefixes gets its own goodness-of-fit p-value, and if the null is true *everywhere*, those 256 numbers are themselves a Uniform$(0, 1)$ sample. They are: a Kolmogorov–Smirnov test — the largest gap between the 256 values' empirical CDF and the uniform's straight line — comes back at $p = 0.62$. Uniformity also prices the scariest number in the table. The worst-fitting prefix scored $p = 0.0015$, damning in isolation; but the *minimum* of 256 uniforms is Beta$(1, 256)$ by the order-statistics law of §4.5, under which $\Pr[\min \le 0.0015] = 1 - (1 - 0.0015)^{256} \approx 0.32$. An ordinary bottom of the class, not a discovery — the arithmetic that separates data dredging from findings.

Origins and habitat. The uniform is classical probability's silent axiom: Laplace's definition of probability as favourable cases over "equally possible" cases (Laplace, 1812) simply *is* the discrete uniform, elevated by his principle of insufficient reason into the default model of ignorance — a move philosophers have argued about ever since (equal in which parameterisation?). Its modern career began at Los Alamos in the late 1940s, when Ulam and von Neumann turned streams of uniform pseudo-random digits into the Monte Carlo method, making the uniform a computational instrument rather than a modeling assumption — the role it plays in every sampler of this library. In data it arises as quantization and rounding error, as the phase at which an unsynchronized observer catches a cycle, and universally through the *probability integral transform*: $F(X) \sim \mathrm{Uniform}(0, 1)$ for any continuous F, the identity behind p-value uniformity, inverse-transform sampling (§2.1), and copula modeling, which separates each variable's margin from the dependence between them.

↪ implemented in `src/dist.c`: `ps_uniform()`

4.2 Exponential

The Exponential(λ) variable (rate λ; density $\lambda e^{-\lambda x}$, mean $1/\lambda$, variance $1/\lambda^2$) is the continuous waiting time to the next event of a constant-rate process — the continuous twin of the geometric, and like it *memoryless*: a component that has survived t hours is statistically as good as new. That property makes it the baseline model of reliability engineering, queueing theory (service and inter-arrival times), and survival analysis, where its constant *hazard rate* is the reference against which ageing (or infant-mortality) effects are measured. The sampler is the one-liner promised in Figure 2: $X = -\ln(1-U)/\lambda$.

Origins and habitat. The exponential earned physical reality around 1900, when Rutherford and Soddy found that radioactive decay follows an exact exponential law with a substance-specific half-life — memorylessness measured in a laboratory. Mathematically it is welded to the Poisson process: exponential inter-arrival times and Poisson counts are two descriptions of the same object, so Erlang's telephone traffic and every queueing model since carry it as the default clock. Like its discrete twin, it is provably the *only* continuous memoryless distribution, and it is closed under competing risks — the minimum of independent exponentials is exponential with the rates summed — which is why systems of many independent failure modes look exponential in their prime of life. Post-war reliability engineering built its handbooks on exactly that constant-hazard middle of the "bathtub curve", and modern survival analysis still starts from the exponential as the null against which ageing (Weibull), frailty, and cure models are judged.

↪ implemented in `src/dist.c`: `ps_exponential()`, `ps_exponential_cdf()`

4.3 Normal

The Normal(μ, σ) density $\frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}$ is the center of gravity of statistics, for one overwhelming reason: the *central limit theorem*. Sums and averages of many small independent effects are approximately normal almost regardless of the ingredients' own distributions — which is why measurement errors, biological traits, and sample means all look Gaussian, and why the normal is the default model for noise in regression and the reference distribution for estimators. Essentially all of §5 exists to make small-sample corrections to normal-based reasoning.

Origins and habitat. No distribution has a richer pedigree. de Moivre (1738) found the curve as the large-n limit of the binomial; Gauss (1809) derived it as the error law under which the arithmetic mean is the best estimate, founding least squares in the process (hence *Gaussian*); Laplace's 1810 central limit theorem explained *why* it appears — sums of many small independent contributions converge to it almost regardless of their individual laws (Laplace, 1812). The nineteenth century then watched it colonise the empirical world: Quetelet fit it to human chest measurements and birthed social statistics' "average man" (1835), Maxwell derived it for molecular velocities (1860), and Galton's quincunx made the mechanism visible in falling beads; the now-standard name "normal" spread through Galton and Pearson late in the century — a naming historians note is unfortunately value-laden. Its natural habitats are exactly the CLT's: measurement errors built from many small disturbances, biological traits summing many genetic and environmental effects, diffusive displacements (Einstein's 1905

Brownian-motion analysis), and averages of almost anything. The perennial caution is in the tails: the CLT speaks about the middle of the distribution, and real-world extremes — finance above all — are routinely heavier than Gaussian, which is one job Student's t (§4.7) takes over.

F^{-1} has no closed form, so inversion is unavailable; instead the library uses the polar variant of the Box–Muller transform (Box and Muller, 1958), our one *rejection* sampler (Figure 7): draw (v_1, v_2) uniformly in the square $[-1, 1]^2$, accept when $s = v_1^2 + v_2^2 < 1$ (probability $\pi/4 \approx 79\%$), and return $v_1\sqrt{-2 \ln s/s}$, which is exactly standard normal.[1]

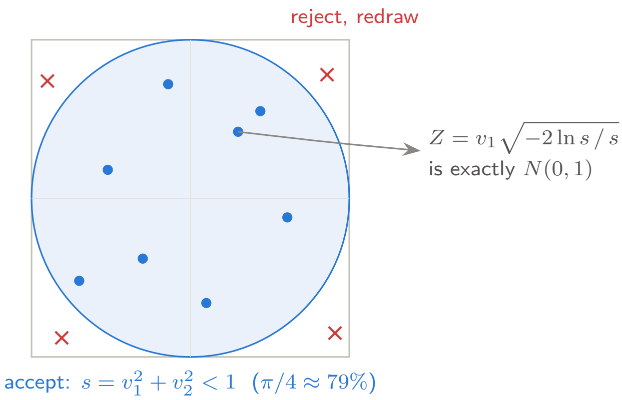

Figure 7: The polar Box–Muller sampler: uniform points in the square are kept only if they land in the disc; the kept point's radius and direction are then reshaped into a standard normal. Rejection turns an easy distribution (uniform on a square) into a hard one (uniform on a disc) at a known, modest cost. [interactive version on the web]

↪ implemented in `src/dist.c`: `ps_normal()`, `ps_normal_pdf()`, `ps_normal_cdf()`

4.4 Gamma

The Gamma(k, θ) family (shape k, scale θ; density $\propto x^{k-1}e^{-x/\theta}$, mean $k\theta$, variance $k\theta^2$) is the flexible law of *accumulated* waiting: for integer k it is exactly the time until the k-th event of a Poisson process — a sum of k exponentials — and general k interpolates smoothly (Figure 8). It models rainfall totals, insurance losses, service times, and neuronal inter-spike intervals; in Bayesian inference it is the conjugate prior for Poisson and exponential rates, so posterior simulation of rate parameters is gamma sampling. It also generates the next two distributions outright.

One law, many hats. Read k as "how much has accumulated" and θ as the unit of accumulation, and the family's special cases stop being coincidences. Exponential(λ) is Gamma$(1, 1/\lambda)$: a single wait. Erlang is Gamma at integer k: the k-th arrival. χ^2_ν is Gamma$(\nu/2, 2)$: an accumulation of ν *half*-waits, one per squared Gaussian dimension — so the dartboard's χ^2_2 (§4.6, Figure 10) is Gamma$(1, 2)$, an exponential in disguise, which is why a dart's squared miss distance is memoryless. The Beta of §4.5 is the *share* $G_1/(G_1+G_2)$ of one gamma in a pair. And as $k \to \infty$ the accumulated sum turns normal — the central limit theorem operating *inside* the family. One law, three hats: waiting time in queueing, accumulated evidence about a rate in Bayesian inference, accumulated squared noise in sampling theory.

[1] The method actually yields *two* independent normals, $v_1\sqrt{-2\ln s/s}$ and $v_2\sqrt{-2\ln s/s}$; `ps_normal()` discards the second so the sampler carries no state between calls. Production libraries chasing speed instead use the ziggurat method (Marsaglia and Tsang, 2000b), a heavily engineered rejection scheme — a good example of the complexity our simplicity rule declines.

A roadside experiment. The family earns its keep — all of it, not just the named special cases — in a place anyone can watch: gaps in traffic. Stand at the roadside with a stopwatch and time the *headways*, the seconds between one car passing and the next. On an empty road at night the arrivals are memoryless and the headways come out exponential — $\mathrm{Gamma}(1, \theta)$, Poisson traffic. In dense traffic the mechanics of car-following take over: no driver sits closer than a safe gap, so the headways bunch tightly around a preferred value — large k, the near-normal end of the family. Ordinary daytime traffic lives in between and fits neither special case, which is why the standard headway model of traffic engineering is the full gamma (Pearson Type III) with fractional shape (May, 1990): the shape parameter acts as a physical *regularity dial* — coefficient of variation $1/\sqrt{k}$ — turning continuously from chaos ($k = 1$) toward clockwork ($k \to \infty$) as the road fills. One phenomenon, one parameter, sweeping the entire family; you can watch rush hour turn the dial. Meteorology learned the same lesson from raindrop sizes: the classical Marshall and Palmer (1948) spectrum was exponential, but it over-counts both drizzle and the largest drops, and the accepted correction (Ulbrich, 1983) is exactly the gamma with its shape set free — a distribution once sampled by hand, by letting drops fall into a pan of flour and sieving the dough pellets they leave.

Summing exponentials only works for integer k, so the library uses the one algorithm where we accept a little cleverness for a lot of practical value: the Marsaglia and Tsang (2000a) method, the standard choice of modern statistical software, and still only a dozen lines. Its idea is a polished version of Figure 7's: push a normal draw Z through the smooth cube $X = d(1 + Z/\sqrt{9d})^3$ with $d = k - \frac{1}{3}$, which lands *almost* exactly on the gamma density, then repair the small discrepancy with a rejection step that accepts well over 95% of proposals. Shapes $k < 1$ (where the density blows up at zero) are handled by a classical boost: draw at shape $k + 1$ and multiply by $U^{1/k}$.

Origins and habitat. The name comes from Euler's gamma function (1729), which normalises the density; as a *statistical* family it entered through Karl Pearson's 1890s system of skew curves (his Type III), built to fit the asymmetric histograms — income, rainfall, biometry — that the normal manifestly could not. Erlang independently arrived at the integer-shape case in 1909 as the duration of k successive telephone holding times, and "Erlang distribution" survives in queueing to this day. The genesis mechanism is accumulation of positive quantities: total rainfall from many showers (the gamma is hydrology's standard wet-day model), aggregate insurance losses, hospital lengths of stay, neuronal inter-spike intervals built from sub-threshold steps. Two structural roles keep it central to inference: it is the variance-component distribution behind §4.6, and in Bayesian statistics it is the conjugate prior for Poisson and exponential rates — a pairing systematised in the conjugate-family framework of Raiffa and Schlaifer (1961) — so that posterior inference about "how often" questions is, computationally, gamma sampling; the gamma-mixed Poisson also yields the negative binomial, the workhorse fix for the overdispersed counts of §3.4.

↪ implemented in `src/dist.c`: `ps_gamma()`, `ps_gamma_cdf()`

4.5 Beta

The $\mathrm{Beta}(a, b)$ family lives on $(0, 1)$ with density $\propto x^{a-1}(1-x)^{b-1}$, mean $a/(a+b)$, and variance $ab/((a+b)^2(a+b+1))$; its shapes (Figure 9) run from uniform through unimodal humps to J- and U-curves. It is *the* distribution for unknown proportions: in Bayesian inference it is the conjugate prior for the Bernoulli/binomial p — start from $\mathrm{Beta}(a, b)$, observe s successes and f failures, and the posterior is $\mathrm{Beta}(a+s,\, b+f)$, which is why Thompson-sampling bandits and Bayesian A/B testing are, computationally, beta simulation. It also models proportions directly (beta regression) and is the order-statistics law: the k-th smallest of n uniforms is

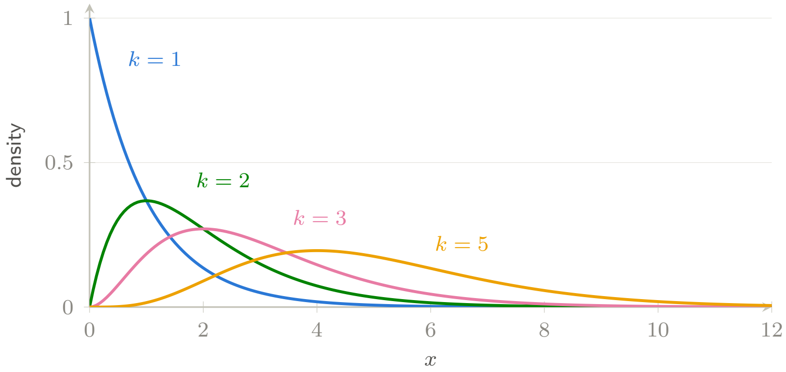

Figure 8: Gamma densities with scale $\theta = 1$. Shape $k = 1$ is the exponential; as k grows the density moves right, symmetrizes, and (by the CLT, since it is a sum of k waits) approaches a normal — the family tree of Figure 1 in a single picture. [interactive version on the web]

Beta(k, $n-k+1$). The sampler uses the classical gamma representation: $X = G_1/(G_1 + G_2)$ with independent $G_1 \sim \text{Gamma}(a, 1)$, $G_2 \sim \text{Gamma}(b, 1)$.

Origins and habitat. The beta is the oldest posterior in statistics. Thomas Bayes's essay (Bayes, 1763), published posthumously by Richard Price, posed the inverse problem — given observed successes and failures, what is the chance itself? — through his billiard-table thought experiment, and the answer he computed is exactly the Beta(s+1, f+1) update this section's demo performs; Laplace, independently, integrated the same distribution to obtain his rule of succession (1774), the first published formula for learning a probability from data. Pearson later absorbed it as Type I of his curve system. It arises naturally three ways: as the law of an unknown proportion, as the k-th order statistic of n uniforms (so ranks, quantile estimates and records lead to it), and as the share $G_1/(G_1+G_2)$ of a gamma-distributed total — whence its multivariate sibling, the Dirichlet. Its application history has a distinctly modern arc: Thompson (1933) proposed allocating treatments by sampling from exactly this posterior, an idea that lay dormant for decades before returning as *Thompson sampling*, the algorithm inside contemporary bandit systems for ad serving, recommender exploration and adaptive clinical trials; the 1950s PERT method of project management adopted the beta for task durations, and empirical Bayes shrinkage of batting averages and hospital rates is Beta–binomial reasoning at scale.

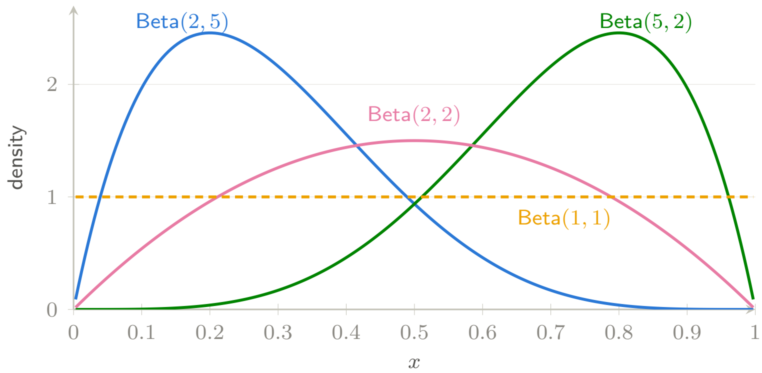

Figure 9: Beta densities: the driver's worked example Beta(2, 5) leans toward small proportions, its mirror Beta(5, 2) toward large ones, Beta(2, 2) is a symmetric hump, and Beta(1, 1) is the uniform — the natural "know nothing" prior for a probability. [interactive version on the web]

↪ implemented in src/dist.c: `ps_beta()`, `ps_beta_cdf()`

4.6 Chi-squared

The χ^2_k variable is a sum of k squared standard normals; it is also exactly Gamma($k/2$, 2), which is how the library both simulates it (one call to `ps_gamma`) and evaluates its CDF. Mean k, variance $2k$. Its starring role is as the *distribution of accumulated squared noise*: in a normal sample of size n, the rescaled sample variance $(n-1)S^2/\sigma^2$ is χ^2_{n-1} — the fact that makes confidence intervals for variances possible and, in the next section, gives the t statistic its denominator. It reappears wherever squared deviations are summed: Pearson's goodness-of-fit and independence tests, and the asymptotic null distribution of likelihood-ratio statistics (Wilks' theorem) — which makes χ^2 tail areas the common currency of model comparison.

Two degrees of freedom: darts and rifles. The cleanest picture of χ^2 is the $k = 2$ case (Figure 10). Throw an arrow at a dartboard, aiming at the bullseye: if the horizontal and vertical misses are independent $N(0, \sigma^2)$ errors X and Y, then

$$\frac{X^2 + Y^2}{\sigma^2} \sim \chi^2_2,$$

which is Gamma$(1, 2)$ — a plain exponential. Its mean is $k = 2$ and its variance $2k = 4$, so the expected squared miss distance is $2\sigma^2$ with standard deviation $2\sigma^2$; the miss *radius* $\sqrt{X^2 + Y^2}$ is the Rayleigh of §4.9. This picture is a working tool at the proof range. To measure a rifle's intrinsic accuracy, the manufacturer clamps it in a machine rest — eliminating the shooter, leaving barrel harmonics, ammunition and manufacturing tolerances — and fires n rounds, recording impacts (x_i, y_i). Two questions, two χ^2 tests. *Is it zeroed?* If the rifle shoots where it points, with design dispersion σ_0 per axis, the group centre obeys $n(\bar{x}^2 + \bar{y}^2)/\sigma_0^2 \sim \chi^2_2$ — the dartboard statistic itself; a large value means systematic bias (a misaligned sight, not bad luck). *Is it built to tolerance?* The scatter about the group centre gives $\sum_i [(x_i - \bar{x})^2 + (y_i - \bar{y})^2]/\sigma_0^2 \sim \chi^2_{2n-2}$ — $2n$ squared deviations minus two estimated means: estimate parameters, lose degrees. With $n = 10$ rounds the statistic is χ^2_{18}, mean 18 and standard deviation $\sqrt{36} = 6$; observe more than the upper percentile $\chi^2_{18,\,0.99} \approx 34.8$ and the dispersion exceeds spec at the 1% level. Because the mean grows as k while the standard deviation grows only as $\sqrt{2k}$, the relative fluctuation $\sqrt{2/k}$ shrinks with every extra round — the arithmetic reason more test shots make a sharper acceptance test.

Three degrees of freedom: a firefly on video. Raise k by one and the dartboard becomes a firefly (Figure 11). Film one flying around a porch light and step through the footage frame by frame: in each frame its position has three independent $N(0, \sigma^2)$ deviations from the bulb — horizontal X, vertical Y, and depth Z, the $k = 3$ dimensions — and the same arithmetic gives

$$\frac{X^2 + Y^2 + Z^2}{\sigma^2} \sim \chi^2_3,$$

now with mean 3 and variance 6. The firefly's distance from the bulb is a scaled χ_3 — Maxwell's law for the speed of a gas molecule, whose square is the scaled χ^2_3 kinetic energy met under *Origins* below: one firefly hovering around a light is, distributionally, one gas molecule's velocity vector.

Balls into bins: measuring deviations from expected counts. The workhorse use of χ^2 in practice is Pearson's own: throw n balls into m bins — n die rolls onto six faces, n PRNG outputs into m equal subintervals, n keys into a hash table's buckets — and ask whether the

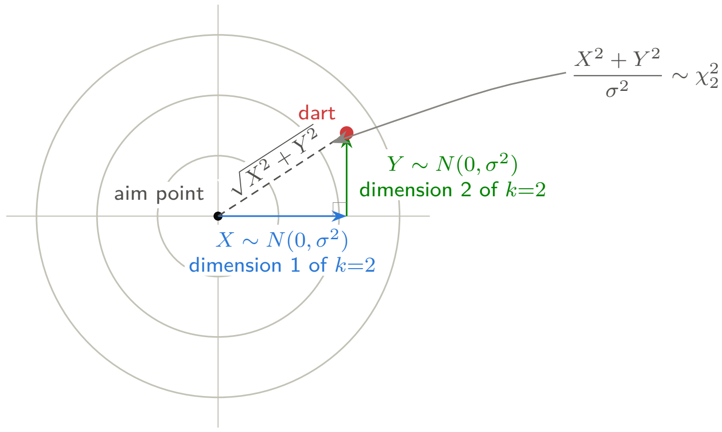

Figure 10: The dartboard geometry of χ^2_2. The dart's horizontal and vertical deviations from the aim point, X and Y, are independent $N(0, \sigma^2)$ errors — they are the $k = 2$ dimensions being summed. Their squared, σ-scaled sum is χ^2_2 (mean 2, variance 4), while the miss *radius* $\sqrt{X^2 + Y^2}$ is the Rayleigh of §4.9. A clamped rifle's proof target is read with exactly this geometry, once per round fired.

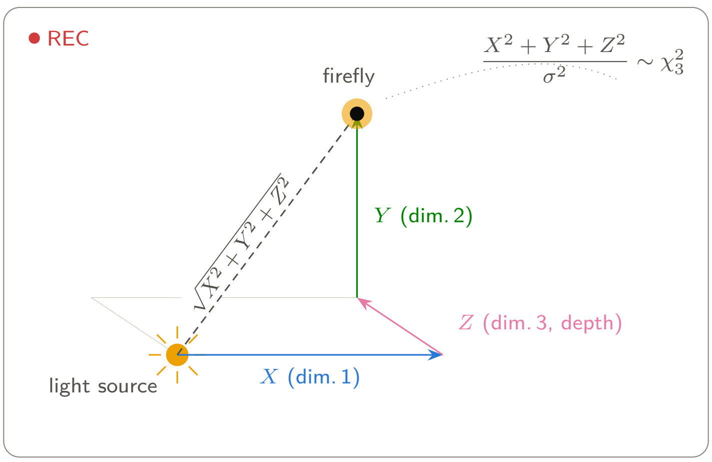

Figure 11: One frame of the firefly video: the $k = 3$ picture of χ^2. The firefly's deviations from the light source — horizontal X, vertical Y, and depth Z — are the three independent $N(0, \sigma^2)$ dimensions of $k = 3$; their squared, σ-scaled sum is χ^2_3 (mean 3, variance 6), and the dashed distance to the bulb is the scaled χ_3 of Maxwell's speed law. The dotted trail is the flight path across earlier frames.

observed occupancies $O_1, \ldots, O_m$ are compatible with the expected ones E_i (all n/m when the null is uniform). The discrepancy measure is

$$Q = \sum_{i=1}^{m} \frac{(O_i - E_i)^2}{E_i} \xrightarrow{d} \chi^2_{m-1},$$

one degree lost because the counts must sum to n, and one more per parameter estimated from the data if the E_i came from a fitted model — Fisher's rule yet again. The analysis in practice has three steps: make the bins fat enough for the asymptotics to hold (the standing rule of thumb is $E_i \geq 5$, merging sparse bins until it does), compute Q, and read the upper tail $1 - F_{\chi^2_{m-1}}(Q)$ as the p-value — in this library, one call to `ps_chi_squared_cdf`. The distribution's moments are the measurement's calibration: under the null Q has mean $m-1$ and variance $2(m-1)$, so the *reduced* statistic $Q/(m-1)$ should sit near $1 \pm \sqrt{2/(m-1)}$ — physicists' "chi-squared per degree of freedom". Values far above 1 mean the counts are too uneven: a loaded die, a biased generator, a hash function clustering its keys. Values far *below* 1 mean they are too even, and are read as evidence of selection or fabrication — Fisher's famous complaint that Mendel's ratios fit better than chance allows. Common measurements built on Q: the goodness-of-fit p-value itself; the effect size $\phi^2 = Q/n$, which unlike Q does not grow with sample size; for an $r \times c$ contingency table (balls classified two ways, expectations from the margins, $(r-1)(c-1)$ degrees) the independence test and its normalized strength, Cramér's $V = \sqrt{Q/(n \min(r-1, c-1))}$; and, run at scale against real generators, the equidistribution half of the Diehard and NIST batteries surveyed by the sibling randtests project. [interactive version on the web]

A field measurement: one χ^2 over 5100 degrees of freedom. The largest balls-into-bins reading in this paper's orbit is the sibling experiment `shastats` (§3.2; The `shastats` project, 2026). Its bins are leading-zero counts (§3.3), pooled from 20 upward so the thinnest expected cell holds about 32 balls — the $E_i \geq 5$ rule observed with room to spare, and fixed *before* looking at the data; its populations are the 256 possible prefix bytes, 2^{25} messages each. The omnibus homogeneity test — one Pearson sum over the 256×21 table, $(256-1)(21-1) = 5100$ degrees of freedom — landed at $\chi^2 = 5139$, $p = 0.35$: a reduced statistic of 1.008 where the calibration above says $1 \pm \sqrt{2/5100} = 1 \pm 0.020$. The prefix does not move the distribution, said with one number. One trap that project dodged is worth recording: its alternative weighting counts each message once per bit string that pads to it, giving a histogram totalling 67 million — but the weights are a deterministic function of the message, so a bin count's variance is $p(1-p)\sum_i m_i^2$, not $p(1-p)\sum_i m_i$, a *design effect* of almost exactly 3. Every χ^2 on the weighted view is divided by that factor (effective sample size 22.4 million, not 67 million) — without the correction, every weighted test screams "significant" at pure bookkeeping. And counts that are too *even* carry information here too, in the Mendel direction just discussed: §5.4 takes up the one under-dispersed reading the sweep produced.

Discrete twin, continuous limit. The binomial has a discrete body and a continuous soul: de Moivre's theorem (§3.2) approximates the integer count by the normal curve. The χ^2 tells the same story from the other shore, twice over. First, it *is* the continuous limit of a discrete variable: Pearson's Q is built from integer counts, so it ranges over a lattice of isolated values — for $m = 2$ bins it collapses to exactly the squared standardized binomial,

$$Q = \frac{(O_1 - np)^2}{np(1-p)}, \qquad O_1 \sim \mathrm{Binomial}(n, p),$$

so $Q \to \chi^2_1$ *is* de Moivre–Laplace, squared. For general m the standardized multinomial deviations live on a lattice inside an $(m-1)$-dimensional hyperplane (the counts must sum to n); the lattice tightens to a Gaussian vector, and the squared length of a Gaussian vector is χ^2_{m-1} by the dartboard logic of Figure 10. The χ^2 is to the multinomial exactly what the normal is to the binomial. The driver runs the convergence live (§6): at 100 balls in 2 bins the lattice is still visible — the nominal-5% test rejects 6.0% of the time — while at 2000 balls in 20 bins the discrete statistic sits flush on its continuous law. Second, the traffic also flows back: every *even-df* χ^2 tail is a finite *discrete* sum,

$$1 - F_{\chi^2_{2k}}(x) = e^{-x/2} \sum_{j=0}^{k-1} \frac{(x/2)^j}{j!} = \Pr[\text{Poisson}(x/2) \le k-1],$$

because $\chi^2_{2k} = \text{Gamma}(k, 2)$ is the waiting time to the k-th event of a Poisson process (§4.4): the k-th arrival comes after time x precisely when at most $k-1$ events have occurred by then. Count and wait are one process described discretely and continuously, and the library leans on the identity in both directions: `ps_poisson_cdf` evaluates the discrete Poisson CDF with one call to the continuous incomplete gamma (§5.1), and the driver prints even-df χ^2 tails beside the Poisson sums they equal — agreeing to all printed digits, because they are the same number.

Origins and habitat. The distribution was first derived by the geodesist Friedrich Helmert (1876) while studying the sample variance of normal observations — surveying error analysis, not statistics, was the motivating problem. Its second birth was Pearson (1900): the goodness-of-fit criterion $\sum(O-E)^2/E$, routinely listed among the most consequential papers in the history of science, because for the first time it let an experimenter ask, quantitatively, "does this model fit these data?" The famous 1922 dispute in which Fisher corrected Pearson's degrees of freedom for contingency tables — estimate parameters, lose degrees — both fixed the practice and forged the modern concept of degrees of freedom. Wilks (1938) then made the χ^2 universal: twice the log likelihood ratio converges to it under the null, so every nested model comparison ultimately cashes out in χ^2 tail areas. In nature it appears wherever squared magnitudes of isotropic noise accumulate: the kinetic energy of an ideal-gas molecule is a scaled χ^2_3 (Maxwell), and the squared length of any k-dimensional Gaussian error vector — radar returns, portfolio residuals — is χ^2_k by definition.

↪ implemented in `src/dist.c`: `ps_chi_squared()`, `ps_chi_squared_cdf()`

4.7 Student's t

Student's t_ν is defined by the construction at the heart of this library:

$$T = \frac{Z}{\sqrt{V/\nu}}, \qquad Z \sim N(0,1), \quad V \sim \chi^2_\nu \text{ independent},$$

and the sampler is that definition verbatim. It looks like a normal but with *heavier tails* (Figure 12): dividing by an estimated, rather than known, scale injects extra uncertainty, and the smaller ν is, the fatter the tails. As $\nu \to \infty$ it converges to the normal; at $\nu = 1$ it is the wild Cauchy with no mean at all. Historically it was discovered by Student (1908) — W. S. Gosset, publishing pseudonymously from the Guinness brewery — precisely because brewery samples were too small for large-sample normal theory. Beyond the t-tests of §5 (and the confidence intervals $\bar{x} \pm t^*_\nu\, s/\sqrt{n}$ dual to them), its heavy tails make it a workhorse *model* in its own right: robust regression replaces normal errors with t errors so outliers stop dragging the fit, and finance uses t innovations because asset returns produce extreme moves far more often than a Gaussian allows.

Origins and habitat. The backstory rewards telling. Gosset worked at Guinness on barley and hops experiments whose samples were unavoidably tiny, and large-sample normal theory visibly failed them; company policy forbade employees to publish (a rival brewery had been embarrassed by a leaked article), so his solution appeared under the pseudonym "Student". The 1908 paper is also an early landmark of *simulation*: to check his curve, Gosset drew 750 samples of size four — by shuffling cards — from Macdonell's anthropometric measurements of 3,000 convicts, a Monte Carlo experiment four decades before the name existed. The result was rigorous but little used until Fisher supplied a clean derivation, extended it to regression coefficients, and built his 1925 handbook around tabulated t (Fisher, 1925), turning small-sample exactness into everyday laboratory practice. Beyond testing, the t arises naturally as a *scale mixture* of normals — a Gaussian whose unknown variance is itself random — which is precisely why it suits populations that are normal-with-heterogeneous-precision: measurement series with occasional sloppy runs, asset returns alternating between calm and turbulent regimes, and any Bayesian analysis where integrating out an unknown variance turns a normal likelihood into a t predictive.

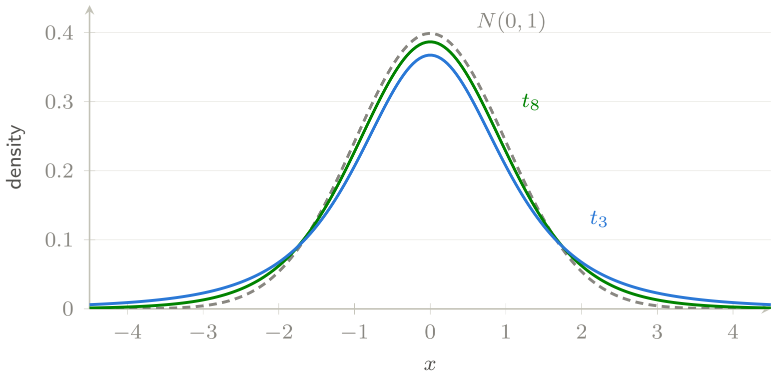

Figure 12: Student's t against the standard normal. The lower the degrees of freedom, the more probability escapes into the tails — so small samples need larger observed effects before a result counts as surprising. This gap between t_ν and $N(0,1)$ *is* the small-sample correction of §5. [interactive version on the web]

↪ implemented in `src/dist.c`: `ps_student_t()`, `ps_student_t_pdf()`, `ps_student_t_cdf()`

4.8 Fisher's F

The F_{d_1,d_2} variable is a ratio of independent normalized chi-squareds, $F = (U/d_1)/(V/d_2)$ — the sampler, once again, is the definition. Since sample variances are scaled chi-squareds (§4.6), F is the natural null distribution for *comparing variances*: its tail areas power the classical ANOVA F-test (is the variance *between* group means larger than the variance *within* groups?), the overall significance test of a regression, and nested-model comparisons. Two connections knit the family shut: $t_\nu^2 = F_{1,\nu}$ (squaring a t-test gives an F-test), and ANOVA with two groups is exactly the pooled t-test.

Origins and habitat. The F is the youngest of the family and the most deliberately engineered. Fisher developed the analysis of variance in the 1920s at Rothamsted Experimental Station, where decades of agricultural field trials — plots of wheat under different manures, varieties and years — demanded a way to split observed variation into components and judge which differences exceeded chance; his 1925 handbook (Fisher, 1925) and the later *Design of Experiments* (1935) wrapped the variance ratio in the trio that still defines good experimentation: randomization, replication, and blocking. Fisher himself worked with $z = \frac{1}{2}\ln F$; George

Snedecor's 1934 tables at Iowa State popularised the untransformed ratio and named it F in Fisher's honour. It arises wherever two independent estimates of noise compete: signal variance against residual variance in ANOVA and regression, a smaller model's leftover error against a larger model's in nested comparisons, and channel power against background in variance-ratio detectors. Its $t^2 = F_{1,\nu}$ identity is the formal statement that the t-test is one-factor ANOVA in miniature — the whole classical testing toolkit is, in the end, ratios of chi-squareds.

↪ implemented in `src/dist.c`: `ps_f()`, `ps_f_cdf()`

4.9 Rayleigh, and the birthday bound

The last two members of our collection are less common in introductory courses but indispensable in one modern habitat: modeling *cryptographic hash functions*. An ideal n-bit hash behaves as a random oracle — each input is a ball thrown uniformly into $N = 2^n$ bins — so its security statistics are the statistics of balls in bins, and those statistics are governed by exactly the distributions of this and the next section.

The Rayleigh(σ) distribution has density, mean and variance

$$f(x) = \frac{x}{\sigma^2} e^{-x^2/2\sigma^2} \; (x \geq 0), \qquad \mathbb{E}X = \sigma\sqrt{\pi/2}, \qquad \operatorname{Var} X = \frac{4-\pi}{2}\,\sigma^2,$$

and two equivalent geneses: it is the length $\sqrt{Z_1^2 + Z_2^2}$ of a two-dimensional Gaussian vector (a scaled χ with 2 df), and it is $\sigma\sqrt{2E}$ for a standard exponential E — the one-line inversion sampler. Its cryptographic starring role is the *birthday problem*: throw balls into N bins until two share a bin, and the waiting time T satisfies $T/\sqrt{N} \to$ Rayleigh(1) as $N \to \infty$. Hence collisions in an ideal n-bit hash appear after about $\sqrt{\pi N/2} \approx 1.25 \cdot 2^{n/2}$ evaluations — the *birthday bound* that halves every hash's collision resistance relative to its output size, dictates that 128-bit hashes are collision-broken at 2^{64} work, and sets the $\sqrt{N}$ running time of Pollard (1975)-style rho methods, whose cycle-finding walks are governed by the same law. The analysis of these collision phenomena is classical (Knuth, 1997, §6.4).

Origins and habitat. Lord Rayleigh derived the distribution in 1880 while asking what amplitude results from superposing many acoustic vibrations of random phase (Rayleigh, 1880) — a two-dimensional random walk in disguise. The same geometry makes it ubiquitous wherever a resultant of many random phasors is measured: multipath radio propagation (*Rayleigh fading*, the baseline channel model of mobile communications), radar clutter amplitudes, sea-wave heights, and wind speeds; and wherever squared Gaussian components pair up — speckle in ultrasound and synthetic-aperture imagery. In the hash-modeling annex it is the distribution to simulate when estimating collision-search cost under resource constraints, as the driver's Rayleigh row does in miniature. The same ball-and-bin picture, run as a test rather than a bound, is Diehard's birthday-spacings test; the sibling survey covers it alongside the rest of Marsaglia's battery.

↪ implemented in `src/dist.c`: `ps_rayleigh()`, `ps_rayleigh_pdf()`, `ps_rayleigh_cdf()`

4.10 Gumbel, and the fullest bucket

Where the Rayleigh governs the *first* coincidence, the Gumbel(μ, β) distribution governs the *worst* one. With CDF and moments

$$F(x) = \exp\!\big(-e^{-(x-\mu)/\beta}\big), \qquad \mathbb{E}X = \mu + \gamma\beta \; (\gamma \approx 0.5772), \qquad \operatorname{Var} X = \frac{\pi^2}{6}\beta^2,$$

it is the extreme-value limit: by the Fisher–Tippett theorem (Fisher and Tippett, 1928), suitably centred maxima of independent light-tailed draws (normal, exponential, gamma, . . .)

converge to it — the CLT's counterpart for maxima rather than sums. The canonical concrete case is exponential draws: $\max(E_1, \ldots, E_n) - \ln n \to \text{Gumbel}(0,1)$, which the web edition's demo simulates live. For hash modeling it answers "how bad is the worst case?": the most-loaded bucket of a hash table, the longest probe sequence, and the longest run of identical bits in an n-bit output stream — approximately $\log_2 n$ plus Gumbel-type fluctuation (up to lattice effects), the null distribution behind the longest-run-of-ones test in NIST's randomness battery for cryptographic generators (Rukhin et al., 2010) — all fifteen NIST tests, and their published criticisms, are walked through on the sibling site. Sampling is double-log inversion, $X = \mu - \beta \ln(-\ln U)$.

A field measurement: the champion digest, and the scariest cell. The sibling experiment `shastats` (§3.2; The `shastats` project, 2026) closes with two textbook extreme-value readings. Each of its populations keeps a champion: the digest with the most leading zeros among 2^{25} tries (§3.3). The champions ranged from 22 to 33 leading zeros, and their distribution matches the *exact* law of the maximum of 2^{25} geometric draws — $F(x)^n$ computed directly rather than the Gumbel limit, because a geometric maximum keeps its lattice wobble, the same caveat the longest-run law above carries — at $p = 0.25$. The second reading resolves the cliffhanger of §3.2: the most lopsided of 6,400 avalanche cells sat +3.9 standard errors from a fair coin. The right yardstick for a *largest* z-score is not the normal but its extreme-value behaviour — location $\sqrt{2 \ln n}$ with Gumbel-scale fluctuation, about 4.35σ expected for the biggest of 6,400 two-sided readings — so the table's scariest cell is in fact slightly *tamer* than randomness predicts. Reporting a maximum without its denominator would have manufactured a finding; pricing it with the extreme-value law dissolved it.

Origins and habitat. Extreme-value theory grew from a practical question — how large a flood must a dam survive? — and Emil Gumbel's *Statistics of Extremes* (Gumbel, 1958) turned the Fisher–Tippett limit into an engineering discipline; annual-maximum river discharges, extreme winds, and design-basis earthquakes are still fitted with his distribution (Gumbel himself was also a courageous public statistician, driven from Germany for documenting political murders). Beyond civil engineering it models record values in athletics and climate, peak electricity demand, and maximal sequence-alignment scores in bioinformatics (BLAST E-values are Gumbel tail areas); machine learning rediscovered it as the *Gumbel-max trick*, which turns softmax sampling into an argmax of Gumbel-perturbed scores. It is the right habitat whenever the quantity reported is a maximum over many comparable chances — which is what worst-case analysis of hashing is.

↪ implemented in `src/dist.c`: `ps_gumbel()`, `ps_gumbel_pdf()`, `ps_gumbel_cdf()`

5 From distributions to inference: the t-test

Everything so far generated data; this section reasons *backwards* from data. The t-test is the classical family's flagship application, and its logic recurs across all of statistics: reduce the data to a statistic whose distribution under a null hypothesis is known exactly (here, thanks to §4.3–§4.7), then ask how extreme the observed value is under that distribution.

5.1 Two special functions carry all the tables

Before software, the answer to "how extreme?" lived in printed tables at the back of textbooks. All of those tables compress into two special functions. The *regularized incomplete beta function*

$$I_x(a,b) = \frac{1}{B(a,b)} \int_0^x u^{a-1}(1-u)^{b-1}\, du, \tag{5.1}$$

is the beta CDF, and the t and F CDFs reduce to it algebraically; for the t,

$$\Pr[T_\nu \le t] \;=\; 1 - \tfrac{1}{2}\, I_{\frac{\nu}{\nu+t^2}}\!\left(\tfrac{\nu}{2}, \tfrac{1}{2}\right) \qquad (t \ge 0), \tag{5.2}$$

with the $t < 0$ case supplied by symmetry, and $\Pr[F_{d_1,d_2} \le x] = I_{d_1x/(d_1x+d_2)}(d_1/2,\, d_2/2)$. Its sibling, the *regularized lower incomplete gamma function* $P(a, x) = \frac{1}{\Gamma(a)} \int_0^x u^{a-1} e^{-u}\, du$, is the gamma CDF and hence the chi-squared and Poisson tail machinery. `probsim` evaluates both with the classical series/continued-fraction pairs (Press et al., 1992, §6.2, §6.4), accurate to about 10^{-14} — each roughly forty lines of C that replace every table in the book.

↪ implemented in `src/special.c`: `ps_incbeta()`, `ps_incgamma_lower()`

5.2 The one-sample t-test

Model n observations as $X_i \sim N(\mu, \sigma^2)$ with both parameters unknown, and test $H_0\colon \mu = \mu_0$. Three facts from §4 assemble the test. Under H_0: (i) $\bar{X} \sim N(\mu_0, \sigma^2/n)$; (ii) $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$, where S^2 is the sample variance with the $n-1$ divisor — exactly the divisor that makes this distributional statement true, which is why `ps_summarize` uses it; (iii) $\bar{X}$ and S^2 are independent. Standardizing $\bar{X}$ and substituting S for the unknown σ produces precisely the $Z/\sqrt{V/\nu}$ construction of §4.7:

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \;\sim\; t_{n-1} \quad \text{under } H_0. \tag{5.3}$$

The unknown σ cancels — that is the miracle Gosset found. The two-sided p-value is then a pure CDF evaluation, $p = 2\, F_{t_{n-1}}(-|t_{\text{obs}}|)$, via equation (5.2):

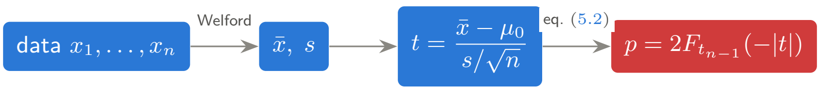

Figure 13 shows the geometry on the library's own test fixture (§6.3): ten measurements with mean 5.47, tested against $\mu_0 = 5$, give $t_{\text{obs}} = 3.12$ on 9 degrees of freedom and $p = 0.012$ — the observed statistic sits well beyond the 5% critical value, so either H_0 is false or a 1-in-80 coincidence occurred. The summaries themselves are computed by Welford's one-pass recurrence (Welford, 1962), which avoids the catastrophic cancellation of the naive sum-of-squares formula.

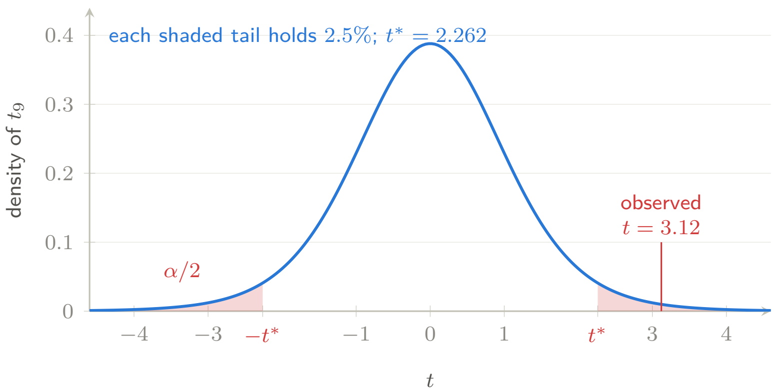

Figure 13: The one-sample t-test on the repository's own test fixture ($n = 10$, df $= 9$). Under H_0 the statistic follows the blue t_9 density; the shaded tails beyond $\pm t^*$ are the 5% rejection region, and the observed $t = 3.12$ lands past them, giving the two-sided p-value 0.012 — the total tail area beyond ± 3.12. [interactive version on the web]

↪ implemented in `src/stats.c`: `ps_summarize()`, `ps_t_test_one_sample()`

5.3 Welch's two-sample t-test

The commonest real question is comparative: do two groups — treatment and control, variant A and variant B — share a mean? With samples $x_1, \dots, x_{n_x}$ and $y_1, \dots, y_{n_y}$, Welch (1947) tests $H_0\colon \mu_X = \mu_Y$ with

$$t = \frac{\bar{x} - \bar{y}}{\sqrt{s_x^2/n_x + s_y^2/n_y}}, \qquad \nu \approx \frac{\left(s_x^2/n_x + s_y^2/n_y\right)^2}{\dfrac{(s_x^2/n_x)^2}{n_x - 1} + \dfrac{(s_y^2/n_y)^2}{n_y - 1}}, \tag{5.4}$$

the degrees of freedom being the Welch–Satterthwaite approximation (Satterthwaite, 1946) — generally fractional, which is why `ps_t_test` carries `df` as a double. Unlike the older pooled test, Welch's does not assume the two variances are equal, and it loses almost nothing when they happen to be; that robustness has made it the default two-sample test in modern practice (and in R's `t.test`). It is the standard analysis for randomized experiments and A/B tests; the same statistic inverted gives the confidence interval for the difference in means that any experiment report should carry.

↪ implemented in `src/stats.c`: `ps_t_test_welch()`

5.4 Testing for overdispersion

The t-tests ask whether a mean has moved. The last test in the library asks something the beta-binomial of §3.6 made precise: whether all the trials really came from *one* coin. Given m clusters, k_i successes in n_i trials, H_0 says every trial everywhere is a flip of the same p; the mixture says each cluster draws its own. Write $\hat{p}$ for the pooled proportion and

$$S = \sum_i \frac{(k_i - n_i\hat{p})^2}{\hat{p}(1-\hat{p})}, \qquad T = \sum_i n_i(n_i - 1), \qquad Z = \frac{S - N}{\sqrt{2T}}, \qquad N = \sum_i n_i. \tag{5.5}$$

S is the usual Pearson-style sum of squared standardized deviations, and the arithmetic behind Z is one line of bookkeeping: $\mathbb{E}[S] = N$ under H_0, while under the mixture $\mathbb{E}[S] = N + \rho T$. So $S - N$ estimates ρT, the moment estimate is $\hat{\rho} = (S - N)/T$, and dividing by the null standard deviation $\sqrt{2T}$ gives Tarone (1979)'s $C(\alpha)$ statistic, standard normal under H_0 and large-positive under overdispersion. The test is one-sided by design: $\rho < 0$ means clusters more uniform than chance allows, which is a different model (and usually a data-entry bug). Because it is a $C(\alpha)$ test, it needs no estimate of ρ under the null and retains good power against the beta-binomial without assuming it — any alternative whose variance inflates linearly in ρ gets caught.

The second form of the statistic is the one to remember:

$$Z = \hat{\rho}\sqrt{T/2} \qquad \Longrightarrow \qquad T \approx 2\left(\frac{z}{\rho}\right)^2. \tag{5.6}$$

Detecting an intraclass correlation ρ takes on the order of $2(z/\rho)^2$ *ordered within-cluster pairs* — and pairs, not trials, are the currency. The cost is quadratic in $1/\rho$, which is why the sex-ratio question of §3.6 settles nothing at ordinary sample sizes: at $\rho = 0.002$ and $z = 3$, equation (5.6) asks for $T \approx 4.5$ million ordered pairs, which in sibships of four ($n_i(n_i - 1) = 12$ each) is some 375,000 families. Five thousand families are not close, a million are comfortable, and that gap — not any disagreement about the data — is most of the published literature. Note also that $T = 0$ when every cluster has a single trial: no within-cluster pairs, nothing to compare, and the test is correctly undefined rather than merely uninformative. §6.2 runs all four cases against data whose truth we set ourselves.

A field measurement: dispersion with the sign flipped. The sibling experiment `shastats` (§3.2; The `shastats` project, 2026) ran exactly this style of audit at scale. Its 256 populations of hashes are 256 clusters; for each leading-zero bin it asked whether the per-prefix counts scatter as a single binomial says they must. A real prefix dependence would be the beta-binomial signature just described — each prefix drawing its own p, variance inflated, dispersion index above one. The measured per-bin dispersion across the 256 populations was 1.00 within noise: the effect is absent by the variance route as well as by the mean route of §4.6. The sweep's one flagged reading ran in the *other* direction — bin 0 of the double hash SHA256d varied *less* across prefixes than chance allows (dispersion index 0.756, two-sided $p = 0.003$) — and the handling is a model of the discipline this section preaches. Under-dispersion is not the alternative being hunted (the text above: $\rho < 0$ "is a different model"); the project had examined 574 quantities, so about 5.7 readings below $p = 0.01$ were *expected* and four were observed; and the oddity was therefore filed as a curiosity worth a second look, not a finding. Knowing which tail of the dispersion scale your alternative lives in — and how many tests you ran — is the whole game.

↪ implemented in `src/stats.c`: `ps_tarone_z()`

6 The driver and the tests

The repository ships two executables that turn the theory above into running evidence: a driver that *shows* the library working, and a test suite that *proves* it.

6.1 Simulation meets theory

The driver (`make run`) simulates 200,000 draws from each distribution at the worked parameters of §3–§4 and prints sample mean and variance beside the theoretical $\mathbb{E}X$ and $\operatorname{Var} X$ — the law of large numbers as a table (excerpt from seed 20260718):

distribution	mean	$\mathbb{E}X$	variance	$\operatorname{Var} X$
Binomial$(20, 0.25)$	5.008	5	3.733	3.75
Poisson(4)	3.990	4	3.990	4
Exponential(0.5)	1.995	2	3.985	4
Normal$(10, 2)$	10.001	10	3.985	4
Gamma$(3, 2)$	6.009	6	12.002	12
Beta$(2, 5)$	0.285	0.286	0.0254	0.0255
Beta-binom$(20, 2, 5)$	5.706	5.714	13.760	13.776
Student's t_8	−0.001	0	1.331	4/3
$F_{5,12}$	1.203	1.2	1.100	1.08
Neg. binom$(3, 0.4)$	4.505	4.5	11.247	11.25
Rayleigh(2)	2.504	2.507	1.710	1.717
Gumbel$(0.5, 2)$	1.648	1.654	6.612	6.580

Between the table and the t-tests the driver runs the balls-into-bins experiment of §4.6 ten thousand times at each of three scales, so the discrete Pearson Q can be watched settling onto its continuous law: at 600 balls in 6 bins the sample mean and variance of Q came out 5.02 and 10.3 against the theoretical 5 and 10, with the nominal-5% test rejecting 5.1% of runs, while the coarsest case — 100 balls in 2 bins — still betrays the underlying lattice with 6.0% rejections. It then prints even-df χ^2 tail areas beside the finite Poisson sums they equal (§4.6); the two columns agree to all fifteen printed digits, because they are the same number.

6.2 The t-tests, live

The driver then runs both tests of §5 on data it has just simulated — a situation where, uniquely, we *know* the truth. A one-sample test of $N(10,2)$ data against its true mean (H_0 true) yielded $p = 0.50$: no false alarm. A Welch test of $N(10,2)$ against $N(11.5,3)$ (H_0 false) yielded $p = 0.020$: difference detected. Re-running with other seeds shows the expected behaviour: under a true null the p-values scatter uniformly (occasionally below 0.05 — type I error is a feature, not a bug), while under the false null they concentrate near zero at this effect size and n.

The driver closes with the experiment of §3.6: simulated sibships of four, either from one common coin or from a coin drawn per family, handed to Tarone's test of §5.4. At the sex-ratio scale of heterogeneity, $\rho = 0.002$, the run prints

simulated truth	$\hat{\rho}$	Z	p
one coin, 10^6 families	-0.00019	-0.45	0.68
$\rho = 0.002$, 5,000 families	-0.00141	-0.24	0.60
$\rho = 0.002$, 10^6 families	$+0.00200$	4.89	5×10^{-7}
$\rho = 0.05$, 5,000 families	$+0.03026$	5.24	8×10^{-8}

which is the sex-ratio disagreement of §3.6 reproduced on demand, with the truth known in advance: the same real heterogeneity is invisible in five thousand families and overwhelming in a million, a homogeneous population never manufactures it, and a ρ twenty-five times larger needs no registry at all. The third and fourth rows are equation (5.6) in action — Z grows as $\rho\sqrt{T}$, so trading a factor of 25 in ρ buys back a factor of 625 in the families required.

6.3 The test suite

`make test` runs about a hundred thousand checks in three layers, in increasing order of subtlety. *Fixtures*: CDF, pmf and t-test results are compared to reference values computed independently with SciPy (each fixture's provenance is recorded beside it in the source). *Identities*: mathematical facts that must hold exactly, such as the symmetry $I_x(a,b)+I_{1-x}(b,a) = 1$ of equation (5.1), or $P(\frac{1}{2}, x) = \operatorname{erf}\sqrt{x}$ linking our incomplete gamma to the C library's `erf`. *Statistical*: with a fixed seed, every sampler's 200,000-draw mean must land within five standard errors of theory; and 2,000 simulated one-sample t-tests under a true null must reject at close to the nominal 5% rate — an end-to-end check that samplers, special functions and test logic agree, and an empirical verification of the uniform-p-value fact of §4.1.

↪ implemented in `app/simulate.c` and `tests/test_probsim.c`

7 A modeler's cheat sheet

The table below compresses the tour into one glance: what each distribution models, and the inference task where it most often appears.

distribution	models	flagship inference role
Bernoulli	single yes/no event	proportion estimation, A/B tests
Binomial	successes in n trials	exact tests of proportions
Geometric	failures until success	retry/attrition modeling
Poisson	rare-event counts	count regression, goodness of fit
Beta-binomial	counts when p varies between units	overdispersion tests; lot sampling
Uniform	total ignorance on a range	p-values under H_0; simulation
Exponential	memoryless waiting time	survival & reliability baselines
Normal	sums of small effects	CLT; regression noise
Gamma	accumulated waits, scales	conjugate prior for rates
Beta	unknown proportions	conjugate prior for p; bandits
χ^2	summed squared noise	variance CIs, GoF, likelihood ratios
Student t	normalized small samples	t-tests, robust models
F	variance ratios	ANOVA, regression F-tests
Neg. binomial	failures before r-th success	overdispersed counts; multi-collision cost
Rayleigh	2-D Gaussian lengths	birthday bound; fading channels
Gumbel	maxima of many draws	worst-case loads; extremes; NIST runs test

For going further, three references span the territory: Devroye (1986) for everything about generating non-uniform random variables, Knuth (1997) for the theory of the uniform generators underneath, and Press et al. (1992) for the numerical special-function craft of §5.1.

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